

Digestate from small-scale biogas plants in central Vietnam produced under mesophilic conditions: friend or foe for local farmers?

Hynek Roubík , **Viktoriiia Chubur**, Department of Sustainable Technologies, Faculty of Tropical AgriSciences, Czech University of Life Sciences Prague, Prague, Czech Republic

Yelizaveta Chernysh, Department of Sustainable Technologies, Faculty of Tropical AgriSciences, Czech University of Life Sciences Prague, Prague, Czech Republic; Department of Ecology and Environmental Protection Technologies, Sumy State University, Sumy, Ukraine

Marek Jelínek, Department of Sustainable Technologies, Faculty of Tropical AgriSciences, Czech University of Life Sciences Prague, Prague, Czech Republic

Le Dinh Phung, Dinh Van Dung, Hue University of Agriculture and Forestry, Hue University, Hue, Vietnam

Van Hau Duong, Jan Banout, Department of Sustainable Technologies, Faculty of Tropical AgriSciences, Czech University of Life Sciences Prague, Prague, Czech Republic

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Abstract: This study evaluates the suitability of digestate from small mesophilic biogas plants in central Vietnam for agricultural application. Digestate samples were collected from 85 farms and analyzed for microbial contamination and elemental composition. The presence of pathogens was examined after anaerobic digestion under mesophilic conditions and the impact of antibiotic resistance in microbes was investigated. Potential integration of solar energy into anaerobic digestion and biogas upgrading to biomethane was also explored. Based on experimental data and prior research, a visualization model was developed to illustrate organic waste decontamination through mesophilic digestion. A block diagram was constructed, integrating various perspectives on digestate utilization to assess the broader environmental impact of biogas technology. © 2025 The Author(s). *Biofuels*, *Bioproducts* and *Biorefining* published by Society of Industrial Chemistry and John Wiley & Sons Ltd.

Supporting information may be found in the online version of this article.

Key words: anaerobic digestion; pathogens; small-scale farmers

Introduction

Major challenges facing developing countries include problematic access to energy resources, slow economic development, and environmental pollution that is threatening health increasingly. An increased demand for energy is driving global attention to renewable energy production, with the aim of increasing it in many countries.^{1,2} Integrating renewable energy sources into power generation using organic waste offers a promising solution to worldwide energy crises, addressing the key need for reliable and sustainable energy sources.^{1,3}

Population growth, urbanization, and economic development are driving a sharp rise in waste generation, particularly in developing countries, with projections showing a 70% global increase by 2050.⁴ Given energy and environmental concerns, finding green, sustainable, and economically viable solutions, including effective waste management strategies, is important for further development.⁵

Effective waste management addresses the waste problem and plays a crucial role in energy production and environmental protection. Human resources are also crucial for the successful adoption of waste management strategies in developing countries.^{6,7} This is particularly evident in Southeast Asian countries, where the effective management of animal waste addresses waste management challenges and provides a crucial opportunity for sustainable energy solutions to be implemented.

The traditional demand for pork meat in Southeast Asian countries is increasing and this is leading to significant amounts of animal manure that require effective management.⁸ The primary livestock products in Vietnam are mainly from the farming of pigs and poultry, with pig farming accounting for the highest share in 2023, with an estimated 4.9 million metric tons produced.⁹ Pig farming is traditionally integral to Vietnam's agricultural sector and is pivotal in the country's economic development, making Vietnam the second largest pork producer in Asia.^{10,11} However, this trend was disrupted by the outbreak of African swine fever in 2019. Productivity in both sectors faced a serious threat, especially in 2020, due to biosecurity challenges related to African swine fever and avian influenza.¹² Larger farms in Vietnam are thought to have comparatively better biosecurity than smaller households, which is significant given the need to improve biosecurity on pig and poultry farms. The government also actively promotes transitioning to larger scale farms.¹²

In recent years, animal production in Vietnam has been changing from small-scale to large-scale concentrated farms. However, the previous lack of proper management of large quantities of pig manure has resulted in an adverse impact on

the environment, including greenhouse gas emissions.¹⁰ Some pig farms directly discharge pig manure into the commune's drainage systems.¹³ This transition is still ongoing and most farms remain small scale but implement appropriate manure-management practices, which facilitate efficient recycling of plant nutrients and generate energy with minimal or no global warming potential, thus reducing the environmental impact. Despite the fact that organic waste might serve as a valuable source of nutrients for plants, without appropriate processing practices it becomes a significant source of pollution.¹⁴ According to Li *et al.*,¹⁵ pig waste exhibited high levels of potentially toxic elements (PTEs).

Waste must be processed and treated properly to improve sustainable animal production.¹⁴ Machete and Chabo¹⁶ outlined the strategies for piggery manure management waste disposal, encompassing various regions across Western, Asian, and African countries, which generally involve the sale of manure, its use in crop cultivation, biogas production, and utilization in fish production. More research is needed to develop a regional-specific framework for integrated manure management that minimizes environmental risks and maximizes profit from agricultural applications.¹⁷ Another source of organic material is human excreta, which has traditionally been applied as a fertilizer despite potential health risks.¹⁸ Improper disposal of raw animal manure and human feces can contaminate the environment with pathogens, potentially leading to infections through direct contact with drinking and bathing water.¹⁹

The application of biogas technology is therefore recommended as it offers an approach to treat waste, neutralize pathogens, and recover resources, which is particularly beneficial in smaller scale applications.²⁰ The anaerobic digestion process mitigates health threats effectively by decontaminating various types of organic waste and destroying pathogens that carry diseases for humans and animals, including crop disease-spreading spores.^{21,22}

Biogas produced through anaerobic digestion (AD) is considered an important waste management technology, which simultaneously produces biogas as a main product and digestate as a byproduct that can be applied as a fertilizer.^{8,23} Anaerobic digestion is the controlled degradation of organic waste in the absence of oxygen and in the presence of anaerobic microorganisms. It is an alternative way to handle animal manure and reduce greenhouse gas emissions.²⁴ The implementation of biogas technology is expected to shift farming communities away from resource exploitation and move them towards increased efficiency through the use of improved technology.²³ Rural households with biogas plants reuse pig manure for energy generation and can reduce fuelwood demand and deforestation significantly, as is the

case in Vietnam.²⁵ Meeks *et al.*²⁶ indicated that there was a significant reduction in fuelwood consumption among Nepalese households using biogas, suggesting potential short-term welfare improvements and positive environmental effects contingent upon broader economic opportunities and forest protection policies.

An additional advantage of anaerobic treatment of animal waste is the reduction in greenhouse gas emissions in comparison with slurry storage.²⁷ Vietnam has committed to a 30% reduction in methane emissions by 2030, with a focus on implementing environmentally friendly solutions to manage livestock waste.²⁸ According to the Netherlands Development Organization (SNV), 41.8% of livestock farms in Vietnam have invested in anaerobic waste treatment methods that produce biogas.²⁹ Approximately 465 370 biogas plants have been constructed in Vietnam, with 450 000 of them small scale, averaging 10 m³ in volume, and the remaining plants ranging from 500 to 2000 m³.²⁹ From 2003 to 2020, six different programs were implemented with the contribution of 290 337 biodigesters, including the Biogas Program in cooperation with the Vietnamese Ministry of Agriculture and Rural Development (MARD) partnered with the SNV.³⁰

One of Vietnam's 2030 Targeted Savings, defined under the E5 Solution, aims to have 5% of rural households using biogas for cooking by 2030, replacing coal and making this technology accessible and beneficial to small-scale farmers as both a means of managing animal manure and an alternative energy source.³¹ The use of digestate as fertilizer in addition to biogas utilization has a minor impact on soil emissions of CO₂ and CH₄.³² According to Ogino *et al.*,³³ integrated cattle manure treatment – which produces biogas, applies digestate to rice paddy fields, and uses some of the rice straw as cattle feed – reduces the environmental effects of eutrophication, energy consumption, and climate change when compared with separate treatment methods. A study by Vu *et al.*³⁴ discussed the effect of combining the liquid phase of the digestate from pig waste with various mixtures of crop residues and biochar. Composting digestate with biochar demonstrated a reduction in emissions in comparison with composting digestate with rice straw.

Biogas technology leads to an improvement in the local environment, and social and technical structures,³⁵ and contributes to increasing basic living standards and an improved quality of life.³⁶ According to Pizarro-Loaiza *et al.*,³⁷ many studies have found environmental benefits from replacing fuel types, reducing methane emissions caused by poor manure waste-management practices, and substituting mineral fertilizers with digestate.

One of the problems with this technology is that positive environmental impacts can be reduced in relation to biogas

leakage and subsequent methane emissions.³⁸ However, a notable benefit is that digestate produced by the organic waste AD process has been identified as a rich source of essential plant nutrients and as a way to improve soil chemical properties;³⁹ however, its safety is a concern for end users. Consequently, ensuring the safety of digestate for agricultural purposes is paramount, with close attention given to heavy metal content and pathogen concentrations to prevent contamination and protect end users. Although livestock and poultry may consume additives containing heavy metals, which are later excreted and can contaminate digestate, the environmental risk of soil pollution should also be considered. This requires further investigation along with consideration of the potential benefits of digestate as an organic fertilizer.³⁹

The aim of this study was therefore to examine digestate from a variety of angles to determine whether local farmers could apply digestate produced under mesophilic conditions from small-scale biogas plants in Vietnam. It also aimed to provide a visual model to illustrate how organic waste is cleaned up by anaerobic digestion in mesophilic conditions while accounting for digestate quality indicators.

Materials and methods

Data collection

The survey was carried out in the Huong Tran and Phong Dien districts located in Thua Thien Hue province in Vietnam. A list of biogas technology owners was provided by the Agricultural Forestry Fishery Extension Center (AFFEC) in Hue city. The farms were stratified across two districts (rural and peri-urban) and by whether or not their toilets are connected to the biogas plant. In total, 85 farms (50 from Huong Tra as the peri-urban area and 35 from Phong Dien as the rural area) were used for data collection and sample collection.

Data collection took the form of semi-structured interviews. Each interview took approximately 0.5 h. A questionnaire was designed to determine the practices of animal husbandry and manure management, the household toilet, as well as the disposal and use of digestate, the feeding of the digesters, and other questions related to biogas technology. Semi-structured interviews were conducted with the head of the household or the person responsible for the biogas plants (mainly the husband or wife).

Sample collection

Digestate was collected from the 85 farms visited. There was a predominant use of fixed-dome biogas models equipped with

compensation tanks (Fig. 1). Specifically, KT1 and KT2 fixed dome digesters were used. KT1 is suited for quality substrate management whereas KT2 is designed for challenging soil excavation and high groundwater or flood conditions. The biogas plants that were analyzed ranged from 6 to 12 m³, with an average size of 6.8 m³. The digesters featured dome-shaped designs with concrete bottoms or brick construction. Both types are detailed by Roubík et al.²³

To ensure representative sampling, the contents of each compensation tank were mixed thoroughly using a sterilized 1 m long blender before sample collection. About 1000 g of samples were collected from different parts of the compensation tank and poured into clean plastic buckets. The samples were collected and kept in an insulated box with cooling elements and transported to the laboratory for bacteriological analysis, which was initiated on the day of sampling. Samples were analyzed for *Escherichia coli* and *Enterococcus* spp.

Measurements of temperature and pH

Temperature probes (EL-USB-1, Lascar Electronics, Salisbury, UK) with a temperature range from −35 to +80 °C and ±1 °C accuracy were used to measure the temperature in the slurry mixing tank.

To measure the pH of the slurry mixture, a pH meter (Meter PCE-228SLUR, PCE Holding GmbH, Germany) was used with a measuring range of 0.00–14.00 pH, a resolution of 0.01 pH, and an accuracy of ±0.02 pH + 2d. The pH meter was calibrated using standard buffer solutions before each measurement. The pH measurements were carried out at regular intervals throughout the experimental period to monitor any fluctuations in the acidity or alkalinity of the slurry.

Enumeration of *E. coli* and *Enterococcus* spp.

For the analysis of *E. coli* and *Enterococcus*, a 10 g digestate subsample was weighed into a sterile stomacher

bag containing 90 mL of Maximum Recovery Dilution (MRD) solution (CM 0733, Oxford, UK). The sample was homogenized at 3000 rpm for 1 min, and a tenfold dilution series was prepared in MRD. Aliquots of 100 µL from each dilution were surface-spread onto Brilliance *E. coli*/coliform Selective Agar (CM1046, Oxoid) and Slanetz and Bartley agar (CM0377, Oxoid). *Escherichia coli* was identified as purple colonies on Brilliance *E. coli*/coliform Selective Agar after 24 h incubation at 37 °C. *Enterococcus* spp. appeared as red-maroon colonies on Slanetz and Bartley agar following a 48 h incubation at 44–45 °C.

Calculation of hydraulic retention time

Hydraulic retention time (HRT) is defined as the period of given quantity of input material that remains in the digester to be acted upon by the anaerobic digestion process. The HRT was calculated based on the estimated input of water and manure input and the digester volume using Eqn (1).

$$HRT = \frac{V_d}{V_i} \text{ (days)} \quad (1)$$

where V_d (m³) is the total volume of the digester and V_i (m³) is the volume of total input added daily. Hydraulic retention time in Vietnam (due to the tropical conditions) ideally varies between 40 and 60 days, depending mainly on the design of the biogas plant and the operating temperature.

Wastewater analysis

A wastewater analysis of the supernatant from the biogas reactor was performed at all 85 biogas plants. Wastewater, specifically the biogas reactor supernatant, was subjected to a comprehensive evaluation and several key indicators were analyzed. These indicators included NO₂-N (nitrite-nitrogen), NO₃-N (nitrate-nitrogen), T-N (total nitrogen), T-P (total phosphorus), BOD₅ (biochemical oxygen demand), SS (suspended solids), and total coliforms. The

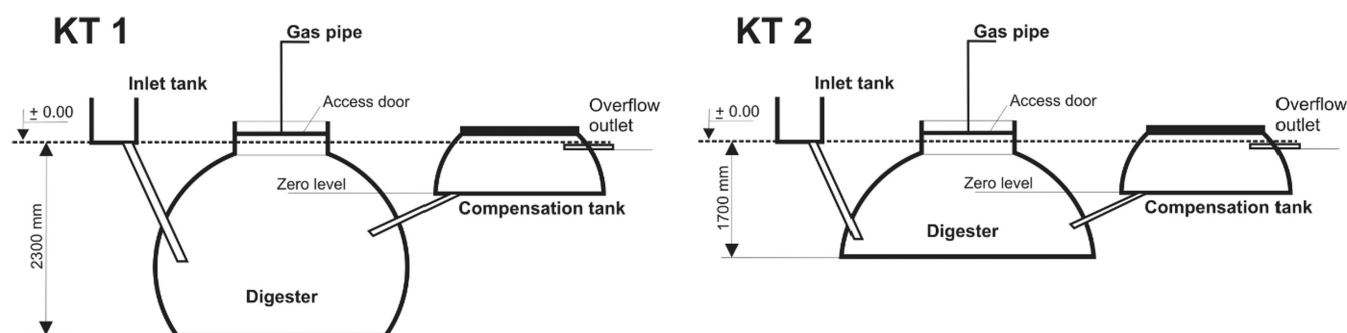


Figure 1. Scheme of small-scale biogas plants used in the target area (adapted from Roubík et al.²³).

analysis was designed to assess the quality and composition of the wastewater in terms of these important parameters.

Wastewater samples were collected from the biogas reactor supernatant in each of the 85 biogas plants. Careful attention was given to the sampling procedure to avoid contamination and to ensure representative samples. Approximately 1000 g of samples were therefore collected from various parts of the thoroughly mixed tanks and poured into clean plastic buckets. Around 200 g of these samples were then transferred to a sterile glass bottle with a screw cap. Samples were stored and transported under cooling conditions to prevent changes in wastewater composition during transport and storage. For the analysis of nitrogenous compounds ($\text{NO}_2\text{-N}$ and $\text{NO}_3\text{-N}$), samples were preserved by adding acid to inhibit microbial activity and stabilize nitrogen species. In the laboratory the wastewater samples collected were subjected to various analytical techniques to determine the concentrations of different indicators.

Analytical measurements for heavy metals detection

Sixty representative samples of digestate were collected randomly, of which 30 were chosen for the detection of heavy metals. To obtain a representative sampling, samples were taken from different parts of a thoroughly mixed tank with a depth of 1 m. Approximately 200 g of the samples were transferred to a sterile glass bottle with a screw cap. The samples were prepared for analysis of heavy metals through homogenization and subsampling to obtain a representative portion for testing. Atomic absorption spectroscopy (AAS) and inductively coupled plasma mass spectrometry (ICP-MS) were used for the analysis of heavy metals and other substances. The substances analyzed in the digestate included Mg, Ca, Al, As, Cd, Cr, Cu, K, Mo, Ni, P, Pb, S, and Zn. They were analyzed using standard procedures, ensuring the accuracy, consistency, and comparability of the findings.

Statistical analysis

The collected data were categorized, coded, and analyzed using the Statistica 10 software package. Comparison of manure management practices (biogas plants with and without connection to the toilet) was made based on χ^2 tests on cross-tabulations. Comparison of the concentration of *E. coli* and *Enterococcus* between farms with and without toilet connections was conducted using linear regression adjusting for clustering within farms. Each factor was tested individually (one by one) and then collectively in multivariate analyses. The presence or absence of *Salmonella* spp. or other pathogens on farms with and without toilet connections was

determined using logistic regression. Hydraulic retention time, pH, and temperature were used as predictors of the reduction in bacterial concentrations. These predictors were assessed individually using linear regression with clustering adjustments at the farm level.

The significance level was set at $P < 0.05$. The value '99' was used in statistical analysis to represent pathogen and indicator concentrations below the method detection limit ($<100 \text{ CFU g}^{-1}$).

Results and discussion

Manure management and toilet practices across owners of small-scale biogas plants

In the developing world, where biogas technology is much less decentralized, small-scale biogas plants are the most common facility of this kind for rural farms. Respondents realize that manure represents a valuable energy and nutritional resource. The common practice among farmers is to use all pig manure as a feed for a biogas plant. This is accomplished by direct connections between the pig pen and digesters. However, in the case of other animals, it varies according to the extent to which manual transfer of manure to the digester occurs. Manure management practices among farmers with and without small-scale biogas plants are described in detail in Roubík *et al.*⁸

The toilet practices of the owners of the biogas plants with a connected toilet were different from those without a toilet connection to the biogas plants. In the case of separate toilets (not connected to the biogas plants) usually kitchen ash is added to the latrine after defecation and most also use lime. There is a belief among users that ash reduces moisture and smell, and lime is viewed as disinfectant for destroying organisms and potential pathogens in the excreta. Similar findings were reported by Jensen *et al.*,⁴⁰ also in Vietnam. However, in the case of toilets connected to biogas plants, lime was not added at all, due to its disinfectant behavior, which would affect the microbiological processes in the biogas plant. Toilets pit latrines without water flushes or pit latrines with manual flushes of approximately 1 L.

Microbiological analysis of digestate

Anaerobic digestion process in relation to the pathogen removal

A diverse range of pathogens is present in organic wastes and manures of both human and animal origin. These include enterobacteria, enteroviruses, parasites, yeasts,

and fungi.¹⁹ Key human and animal pathogens include *Salmonella* spp., *E. coli* (including toxigenic forms), *Vibrio cholera*, *Campylobacter* spp., *Listeria monocytogenes*, *Yersinia enterocolitica*, *Staphylococcus* spp., *Cryptosporidium*, *Mycobacterium*, *Clostridium*, *Campylobacter*, *Giardia*, and *Toxoplasma* and parasites such as *Ascaris* spp., *Taenia* spp. and protozoan parasites, and viruses such as hepatitis E and some exotic viruses like the Nipah virus.^{19,41–43} Many of these are zoonoses, meaning they can be transmitted between animal and human populations, often through the consumption of contaminated freshwater plants.^{43,44} These pathogens are particularly prevalent among human and animal populations in developing countries and therefore often found in fecal material.^{19,43} Their persistence in the environment is affected by local climate, temperature, time spent in the reactor (hydraulic retention time), competitive microbial interactions, pH, soil type, animal host prevalence, topography, land cover, land management, organic waste, operational conditions, as well as their antimicrobial resistance and the presence or buildup of toxic materials.^{19,41,45–49} As human exposure to pathogens is highly undesirable and can easily be linked to contamination levels around the farm or home on the farm and in local water courses, it is essential to prevent and minimize such exposures through adequate pathogen prevention and removal methods.

Anaerobic digestion is being used increasingly in developing countries for the management of organic waste. In central Vietnam, digestion is typically done at mesophilic temperatures (35–40 °C), or exceptionally at thermophilic temperatures (50–55 °C) if the system is heated. Mesophilic digestion is, however, associated with limited pathogen reduction in contrast to thermophilic digestion, where, due to high temperatures, the majority of pathogens are inactivated.^{41,50}

Anaerobic digestion, as a synergic effect of a consortium of microbes, can be classified along with a series of metabolic pathways. However, while the microbial kinetics of anaerobic processes are well understood and described (for example, studies by Nakasaki *et al.*⁵¹, Jain *et al.*⁵² or Kinyua *et al.*⁵³), there is still relatively little information on the processes of pathogen reduction through anaerobic digestion in small-scale biogas plants. In developing countries in general (where applications are small scale), the stability and efficiency of the process are likely to be lower; an awareness of the microbiological safety of digestate and treated sludge is therefore essential as it has implications for human health and pathogen transmission in the food chain and environment.¹⁹

Pathogens can survive the anaerobic digestion process (see Table 1) and therefore persist in the digestate.⁶² In general,

Table 1. Review of presence of pathogens in digestate after anaerobic digestion in mesophilic conditions.

Review of presence of pathogens in digestate after AD treatment in mesophilic conditions (continuous digestion process)			
Organism	Temperature	Proved presence in digestate after AD treatment (yes-y/no-n)	Reference
Bacteria			
Total coliforms	24 (20–26.5)	y	[42]
Fecal coliforms	24 (20–26.5)	y	[42]
<i>Escherichia coli</i>	25 (20.5–29.5)	y	[41]
	24 (20–26.5)	y	[42]
	30	n	[54]
	35	n	[55]
<i>Salmonella</i> spp.	25 (20.5–29.5)	y	[41]
	24 (20–26.5)	y	[42]
<i>Shigella dysenteriae</i>		n	[56]
<i>Enterobacteriaceae</i>		y	
<i>Enterococcus</i> spp.	25 (20.5–29.5)	y	[64]
	25 (20.5–29.5)	y	[41,57]
<i>Campylobacter jejuni</i>	24 (20–26.5)	y	[42]
<i>Yersinia enterocolitica</i>	24 (20–26.5)	y	[42]
<i>Clostridium perfringens</i>	25 (20.5–29.5)	y	[41]
	24 (20–26.5)	y	[42]
	37	y	[58]
	35	y	[57]
	38	y	[55]
	28–35		[59]
Rine norovirus	37	N/A	[60]
<i>Cryptosporidium parvum</i>	37	y	[58]
<i>Mycobacterium</i> spp.	37	y	[58]
<i>Listeria monocytogenes</i>	37	y	[58]
<i>Bacillus cereus</i>	35	y	[59]
	30	y	[61]

hygienization of the end product depends on various factors such as quality of substrate, digester performance (digestion temperature, HRT, pH, ammonium concentration), and pretreatment or post-treatment such as composting.^{62,63}

For comparison, European legislation classifies animal by-products into three categories.⁶⁵ Category I consists of high-risk material, such as substrate from sick animals, which

must be incinerated and not used for biological treatment. Category II includes slaughterhouse waste and manure, which must be pasteurized at 70 °C for 1 h prior to digestion, although manure may be digested without pretreatment if it contains no – or low concentrations of – *Salmonella*, *Enterococaceae*, or *E. coli*, according to Commission regulations.⁶⁶ Category III, covering organic household waste and food processing waste, does not require pretreatment before digestion.⁶⁵ However, this regulatory framework is unsuitable for rural small-scale biogas plants in Vietnam, where pretreatment is not typically applied.

Under mesophilic operating conditions (35 °C) in the small-scale biogas plant, the mean values of the time to 90% reduction in bacterial concentration were 1.8 days for *E. coli*, 2.0 days for *Salmonella dublin*, and 0.9 days for *Staphylococcus aureus*. Under thermophilic conditions (53 °C), Seruga et al.⁶⁷ determined the mean times as 0.4 h for *E. coli*, 0.6 h for *Salmonella dublin*, and 0.5 h for *Staphylococcus aureus*. In general, thermophilic digestion has advantages over mesophilic digestion in pathogen inactivation but, due to energy consumption, process stability, and diversity of anaerobic microorganisms, mesophilic digestion is more widely used in practice. This was confirmed by previous studies.^{59,67} Given these factors, further innovations in pathogen inactivation are needed for the anaerobic digestion process.

Pathogen mitigation via small-scale biogas plants

The World Health Organization has stated that manure has a high concentration of worm eggs, weed seeds, and over 150 different species of animal and human pathogens. By penetrating ground and surface waters, manure and dung runoff degrade water quality dramatically. As mentioned earlier, digestate is formed during anaerobic fermentation of biomass (agricultural waste), which initially contains microbial flora, eggs, and larvae of helminths, and seeds of weed plants. In the mesophilic (36 °C) and thermophilic (54 °C) digestion modes, the substrate inside the reactor heats up, which in turn contributes to the death of parasites and weeds.

Table 2 presents an analysis of pathogenic microflora, decontamination efficiency, and the presence of helminth eggs and weed seeds in organic fertilizers.

The initial microbial load of the manure was 10⁹ CFU. After anaerobic digestion in a biogas plant, it decreased to 10⁷ CFU, indicating a 99% reduction in microbial contamination. The final organic fertilizer contained no helminth eggs, and all weed seeds had lost viability.

Many studies have also reported the death of around 90% of animal pathogens in anaerobic systems, significant or complete destruction of plant pathogens under mesophilic conditions, and the absence of helminth eggs.^{57,59} Johansen et al.⁶⁸ showed that anaerobic digestion under mesophilic conditions effectively reduced the germination of weed seeds (*Brassica napus*, *Avena fatua*, *Sinapsis arvensis*, *Fallopia convolvulus*, *Amzincia micranta*, *Chenopodium album*, and *Solidago canadensis*) and ensured their complete death on day 11 of fermentation. It was found that, under thermophilic conditions, the death rate was faster than under mesophilic conditions.

Death rates of log reductions over a given time period can be used to assess the effectiveness of a treatment process for removing pathogens.¹⁹ Coté et al.⁶⁹ studied the efficiency of a low-temperature anaerobic laboratory-scale digester in reducing viable populations of indicator microorganisms (total coliforms, *Escherichia coli*) and selected pathogens (*Salmonella* spp., *Yersinia enterocolitica*, *Cryptosporidium*, and *Giardia*) in swine slurries operated at 20 °C during 20 day cycles, where it reduced total coliforms by 1.6 to 4.2 log₁₀ units (CFU g⁻¹) (97.9%–100%), and *Escherichia coli* by 2.5–4.2 log₁₀ units (99.7%–100%). *Salmonella* spp. decreased to undetectable levels (<100 CFU g⁻¹).

Oshoma and Omogbai investigated the reduction of viable numbers of *Salmonella* and coliform bacteria during anaerobic and aerobic digestion of cow manure in a laboratory study.⁷⁰ Anaerobic digestion without heating, with a final pH 8.1, reduced viable numbers of *Salmonella* and coliform bacteria by 1.05 × 10⁴ and 1.26 × 10⁴ cells·mL⁻¹, respectively, exceeding aerobic digestion reductions (4.50 × 10⁶ and 1.58 × 10⁶ cells·mL⁻¹). Lowering the temperature from 31 ± 1 °C to 26 ± 1 °C resulted in increased

Table 2. Analysis of pathogenic microflora in organic fertilizers, decontamination efficiency, and the presence of helminth eggs and weed seeds.

	Bacteria contamination, colony forming unit (CFU)	Titer	Disinfection efficiency, %		
			Bacteria contamination	Helminth eggs, pcs·dm ⁻³	Weed seeds, pcs·dm ⁻³
Aerobic treatment	10 ⁹ CFU	3 × 10 ⁵ CFU	–	Presence	Presence
Anaerobic digestion in mesophilic conditions	10 ⁷ CFU	3 × 10 ³ CFU	99	Absence	Absence

viable cell counts in the filtrates. The reduction of *Salmonella* spp. suggests anaerobic digestion has potential for controlling infectious diseases on farms. Moreover, the study of Seruga et al.⁶⁷ confirmed the high sanitary effect of anaerobic co-digestion of food waste. The bacteria *Salmonella Senftenberg* W775 and *Enterococcus* spp., and *Ascaris suum* eggs were eliminated or inactivated during co-digestion of restaurant and organic fraction of municipal waste.

Seruga et al.⁶⁷ suggested that anaerobic digestion under thermophilic conditions can produce a microbiologically safe product, even with heavily contaminated raw materials. In compliance with legal requirements. The inactivation of pathogens studied was achieved within 6.06, 5.5, and 10 h, respectively, demonstrating a rapid process that prevents environmental contamination by pathogens after digestate application.

However, a study by Fu et al.⁷¹ determined the presence of viable but nonculturable (VBNC) *Salmonella* spp. and *Shigella* spp. bacteria in biosolids during anaerobic digestion and their reactivation during cake storage. Thermophilic anaerobic digestion tends to induce pathogens like *Salmonella* spp. and *Shigella* spp. into a VBNC state, rather than eliminating them completely. This suggests that the biological safety of sewage sludge from temperature-phase anaerobic digestion may be compromised, requiring careful evaluation.⁷¹

The presence of pathogens in municipal waste sludge poses a significant risk of waterborne disease outbreaks when used for agricultural purposes. Effective pathogen reduction requires specific conditions, such as acidic environments or controlled pH levels, as demonstrated by Fukushi et al.⁷²

Anaerobic digestion can achieve these conditions. *Salmonella* spp. were shown to be eliminated within 2 days in a sludge-organic acid mixture simulating an acidic digester at a pH below 5.5 under mesophilic conditions (37 °C).⁷²

The digestate can be aerated for several days, aerobically composted, or processed to biochar before being used as a fertilizer. These procedures are crucial for the removal of previously preserved veterinary antibiotics (VAs). However, some may still persist as a result of the nature of VAs. Optimizing VA parameters such as temperature and solid retention time can lead to changes in the structures of the microbial community and improve its ability to biodegrade organic manure contaminants. For example, high-temperature pretreatment or microwave pretreatment combined with mesophilic anaerobic digestion has shown significant improvements in the removal of manure contaminants. Finally, the addition of controlled composting after the application of manure may be necessary to further reduce the levels of manure contaminants to levels that are no longer hazardous when manure is applied to the soil for crop fertilization.⁷³

Figure 2 shows possible directions for reducing antibiotics in digestate.

Based on our research and previous studies, we formed a general model of pathogen reduction (Fig. 3) through the use of small-scale biogas plants under mesophilic conditions.

Liquid phase digestate analysis

Although direct application of digestate to soil is widely used, the large amount of digestate produced in regions

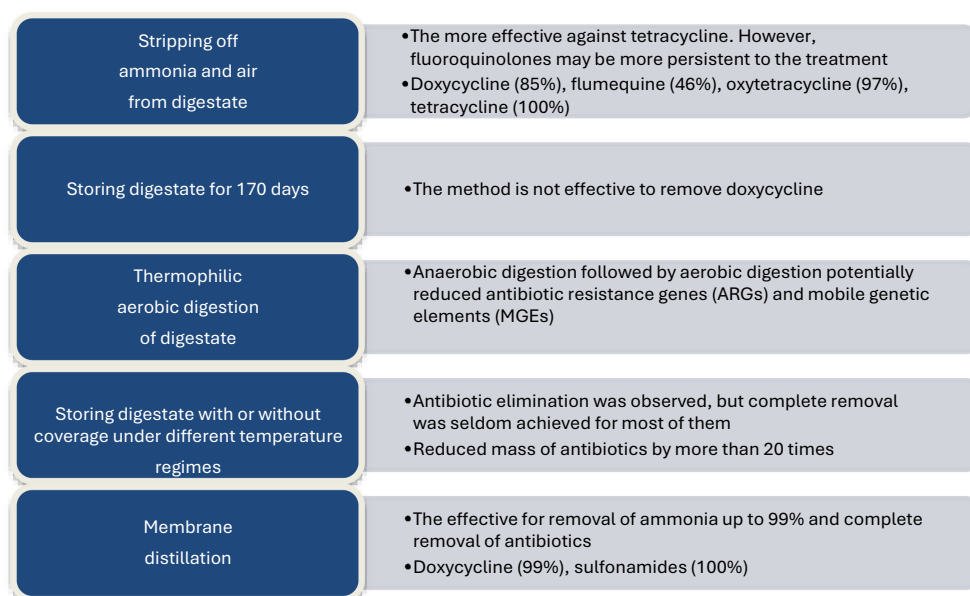


Figure 2. Approaches to removing antibiotics from digestate. Based on Uluseker et al.⁷⁴

with intensive livestock farming can cause problems with nutrient surplus. The high water content (>80%) in digestate increases transportation costs and complicates disposal. To avoid negative environmental impacts, additional digestate treatment is necessary in the context of removing and/or recovering excess nutrients.⁷⁵ Separation of the solid and liquid phases yields two fractions of materials that can be treated independently.⁷⁶

The liquid fraction can be returned to the process, treated to meet land requirements (nutrient removal), or recycled to extract valuable nutrients (P and N).⁷⁷ Common solid–liquid separation technologies are mechanical and chemical.⁷⁸ The type of technology used in solid–liquid separation will determine the composition of the fractions. Moreover, separation efficiency is determined by the flows, solids content, use of additives, and digestate decomposition.⁷⁹

In solid–liquid separation, the liquid fraction is more interesting because it contains a higher proportion of nitrogen and potassium. The digestate liquid fraction can be used to dilute high-solids feedstock and re-feed the digester and/or it can be used as irrigation water.⁸⁰ However, N accumulation due to liquid recirculation can lead to anaerobic inhibition of biomass⁸¹ and limits land-use directives restricting the spread of N on land.⁸²

The experimental data presented in Table 3 are a summary of the analyzed samples, which can be found in Supporting

Information, Appendix S1 (Supporting Information, Table S1). The main emphasis is on the presence of ammonium in the calculation of nitrogen compounds. The aim of nitrogen compound removal is to reduce nitrogen concentration in digestate, convert ammonium to gaseous N, and increase N concentration to regulatory limits or land requirements; the aim of recovery is to fix and concentrate ammonium in a liquid or solid medium potentially suitable for reuse as an agricultural fertilizer or chemical reagent.⁷⁶

Digestate contains a considerable amount of nitrogen in the forms $\text{NO}_2\text{-N}$, $\text{NO}_3\text{-N}$, and $\text{NH}_4\text{-N}$. As can be seen from the results in Table 3, most of the N in the digestate is in the form of ammonium ($\text{NH}_4\text{-N}$), which makes it easily available to plants but can also cause N losses through evaporation in the form of NH_3 . The amount of nitrogen present as nitrite ($\text{NO}_2\text{-N}$) was measured and found to be low (no more than $0.53\text{ mg}\cdot\text{L}^{-1}$, most frequent value range $0.01\text{--}0.05\text{ mg}\cdot\text{L}^{-1}$ – Table 3). Nitrite is an intermediate so it will be consumed and rapidly converted to nitrate ($\text{NO}_3\text{-N}$) under most nitrification conditions. The biochemical oxygen demand over 5 days (BOD_5) was also measured as an additional indicator of biological activity (Table 3). The overall decrease in soluble BOD_5 over the cycle indicates significant heterotrophic bacterial activity. Such activity can degrade insoluble compounds that survive anaerobic digestion. Based on the BOD concentration of the digestate liquid fraction from the

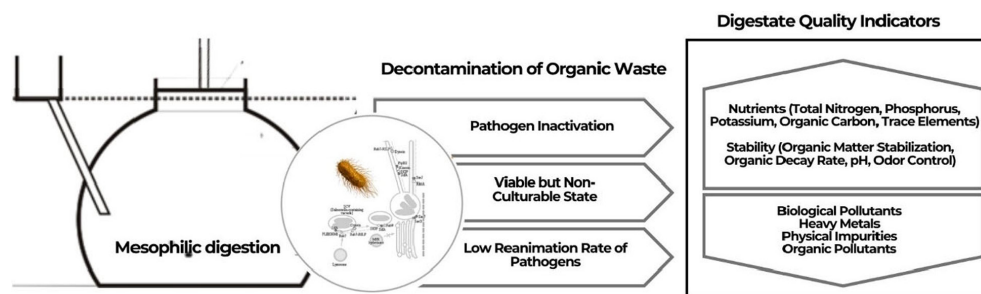


Figure 3. Decontamination of organic waste during anaerobic digestion under mesophilic conditions.

Table 3. Parameters of liquid phase digestate analysis.

Parameter	$\text{NO}_2\text{-N}$	$\text{NO}_3\text{-N}$	$\text{NH}_4\text{-N}$	TN	TP	BOD_5	SS	Total coliforms
Unit	$\text{mg N}\cdot\text{L}^{-1}$	$\text{mg N}\cdot\text{L}^{-1}$	$\text{mg N}\cdot\text{L}^{-1}$	$\text{mg N}\cdot\text{L}^{-1}$	$\text{mg P}\cdot\text{L}^{-1}$	$\text{mg O}_2\cdot\text{L}^{-1}$	$\text{mg}\cdot\text{L}^{-1}$	$\text{MPN}\cdot 100\text{ mL}^{-1}$
Average	0.06	0.55	230.83	481.65	63.92	230.82	2895.95	54.4×10^8
Min.	0.01	0.10	53.70	81.90	15.60	68.00	33.00	2.9×10^3
Max.	0.53	1.70	564.60	1120.90	183.50	597.00	64 167.00	110×10^9
Frequent value range (Q1–Q3)	0.01–0.05	0.2–1.0	142.8–202.0	331.8–579.7	40.1–87.3	154.0–274.0	123.0–519.0	$21 \times 10^7\text{--}110 \times 10^7$
<i>Limit of detection</i>								
$\text{NO}_2\text{-N} \geq$	0.005	$\text{mg}\cdot\text{L}^{-1}$						
$\text{NO}_3\text{-N} \geq$	0.05	$\text{mg}\cdot\text{L}^{-1}$						

operating biogas digesters it was estimated that approximately 80% of the soluble BOD₅ was removed on average. This result was in accordance with other data.⁸³

Numerous studies have examined the release of large amounts of ammonia, which is the primary issue with some anaerobic digestion processes.^{84,85} Ammonium ions (NH₄⁺) and free ammonia (NH₃) are the two major forms of inorganic N present in equilibrium. Free ammonia can pass through the cell membranes of microorganisms, causing intracellular imbalance. The concentrations at which ammonia inhibits microorganisms vary depending on several parameters such as pH, temperature, alteration of the C/N ratio, substrate composition, and origin of the inoculum.⁸⁶ Provolo *et al.*⁸⁷ found that the nitrogen recovery rate from a liquid digestate distillation process was higher than that of the solid fraction.

The ammoniacal nitrogen content hinders the ability to concentrate these digestates by evaporation because it tends to release ammonia gas (NH₃), which reduces the quality of fertilizer and also creates an air pollution problem. Untreated digestate generally has a pH of 7.5 to 8.5, indicating the presence of significant concentrations of the nonionized form of NH₃. The use of ammonia as the sole source of nitrogen for plants is also not ideal as it can lead to acidification of the root zone of the plant. From this point of view, digestate nitrification appears to be a logical step towards stabilizing nitrogen and making digestate suitable for evaporative concentration.⁸³ Wijesinghe⁸⁸ investigated the addition of clay mineral zeolite to slurry or digestate in an anaerobic digester to reduce NH₄⁺ concentration and thus mitigate ammoniacal inhibition. However, the effects of zeolite physicochemical properties on the reduction of total ammoniacal nitrogen in digestate and the mechanisms for improving biogas production should be clarified. The recovery of N from swine manure is also an option to reuse N in agriculture and reduce the amount released into the environment.⁸⁸

As indicated by Lin *et al.*,⁸⁹ about 90% of P in manure is inorganic and, in general, the ratio of inorganic to organic matter does not change significantly after digestion: in digestate samples, inorganic P accounts for 86% of total phosphorus. Such qualitative findings are consistent with our results (Table 3).

Two perspectives exist regarding the impact of anaerobic digestion on P availability: one suggests that the process enhances plant nutrient availability whereas the other asserts that it has no direct effect. Increased pH shifts the chemical equilibrium towards the formation of dissociated phosphate ions, promoting their precipitation as insoluble calcium and magnesium phosphates. Anaerobic digestion may also regulate the binding forms of other elements, such

as Fe. During anaerobic digestion, a portion of dissolved P mineralizes and binds to suspended solids, leading to a significant reduction in the water-extractable P fraction.⁸⁹

The reduction of fecal coliforms was evaluated (Table 3). As the solids content is reduced by digestion, the reported reduction in fecal coliforms was significantly greater than if reported on a mass basis. A conservative estimate of an expected log fecal coliforms reduction of no more than 1.5 log seems appropriate in light of all the available data, and this estimate is consistent with the findings of a study conducted by other authors.⁹⁰ Farrell and Schafer⁹⁰ substantiated the most effective method to increase the reduction of fecal coliform bacteria is to use two (or more) digesters in series with conversion to a pull-and-fill feeding method once a day, or to follow a one-stage breakdown with anaerobic storage.

Heavy metals in digestate

In this paper, the heavy metal content of the mesophilic-degradable biogas digestate was investigated. Table 4 lists the final heavy metal concentrations in the biogas digestate. The average concentrations are compared with limits required by EU standards.

Table 4 and Supporting Information, Appendix S1 (Supporting Information, Table S3) show that the limits are mainly exceeded for Zn and Cu. It is necessary to conduct additional research on the forms of the chemical bonds of these metals, which affect their uptake by plants and the dosage of digestate when applied in agriculture.

Zinc affects the assimilation processes of mineral nutrients: absorption by the root system and transport to the above-ground plant organs. It enhances the absorption of B and Cu but reduces the intake of K, Fe, and Mn, as well as Cd and Pb, and regulates P metabolism.

Zinc has a role in auxin formation and plant growth rates. By increasing the activity of tryptophan synthetase, Zn determines the synthesis of the amino acid tryptophan (a precursor of auxin). The role of Zn in protein synthesis and RNase activity has been established. It is a component of many enzymes, plays an important role in redox reactions, and participates in the regulation of ATP formation.

As Rudaniti *et al.*⁹¹ stated, the optimum Zn spray rate to achieve a significant grain yield response was in the range of 1.0 to 1.5 kg Zn·ha⁻¹. The low bioavailability of Zn in soil can potentially damage plants, humans, and animals.

A study by Tran *et al.*⁹² demonstrated the effect of the trace elements Cu and Zn, which play an important role in plant growth and development, on the accumulation of lead from the soil contaminated with the vegetables studied. The results showed that high concentrations of Cu and Zn in soil cause

Table 4. Content of heavy metals in digestate from mesophilic conditions (N = 30).

	Al	As	Cd	Cr	Cu	Mo	Ni	Pb	S	Zn
	mg·kg ⁻¹	mg·kg ⁻¹	mg·kg ⁻¹	mg·kg ⁻¹	mg·kg ⁻¹	mg·kg ⁻¹	mg·kg ⁻¹	mg·kg ⁻¹	mg·kg ⁻¹	mg·kg ⁻¹
Average	9207.6	10.4	1.5	51.3	1522.1	13.9	15.8	20.5	10 688.1	3854.9
Min	1960.0	2.9	0.3	8.2	241.0	1.7	4.8	2.5	3380.0	758.0
Max	85 235.0	33.1	12.6	337.3	19 195.0	230.2	223.5	166.4	114 094.0	53 490.0
Frequent value range (Q1–Q3)	5192–8208	5.56–13.26	0.79–1.26	11.8–21.2	579–1240	5.45–7.74	6.97–10.16	6.6–22.2	6203–8366	969–2222
Limit			2	600	600		100	300		1500
Cases over limit (%)			3 (9%)	0	21 (87%)		1 (3%)	0		18 (54%)

Note: Bold values represent limits for the respective parameters. These values indicate the threshold above which a measurement is considered exceeding the acceptable standard. In the “Cases over limit (%)” row, the numeric values indicate the number of cases exceeding the limit, while the values in parentheses represent the percentage of cases exceeding the defined limit.

competition with Pb in the process of uptake and accumulation in the plant. In particular, Cu and Zn had an inhibitory effect on Pb uptake and accumulation by these plants.⁹²

Nutrient translocation in spinach and plant growth were mainly negatively affected by Cu in leaves, and the overall effects of Cu in leaves on nutrients in roots and leaves, nutrient translocation in spinach and plant growth were the highest. The results of Liu *et al.*⁹³ showed that the phytotoxicity of Cu, including inhibition of spinach growth and mineral disturbance in spinach, was mainly affected by Cu concentrations in leaves.⁹³ As Zhao *et al.* reported,⁹⁴ *Phytolacca americana* supplemented with increased concentrations of Zn and Cu showed decreased biomass production and stunted plant growth. When the highest concentrations of Zn (2000 mol·L⁻¹) and Cu (100 mol·L⁻¹) were added to plants, toxicity symptoms corresponded to a decrease in dismutase (SOD) or catalase (CAT) activity and an increase in ascorbate peroxidase (APX) and guaiacol peroxidase (GPX) activity.

Studies by Tran *et al.*⁹² showed that Cu and Zn inhibited the entry of lead from soil into the biomass of lettuce, spinach, carrots, and potatoes. Thus, the presence of zinc in the soil may increase the amount of free lead in the soil fluid as a result of competition for the adsorption site on the soil particles, releasing some lead as adsorbent in the soil colloid. This may increase the uptake of lead by plants but competition between Pb and Zn for uptake and transportation from soil to plants can reduce the accumulation of Pb in biomass. It is therefore feasible to employ micronutrients in agriculture to lessen the accumulation of heavy metals (such as Pb and Cd) from soil to plants to maintain the safety of food production.⁹²

In some cases high levels of Cd and Ni were detected that require further monitoring (Table 3). Potential sources of

heavy metals, including Cd and Ni, may vary. However, analysis of input substrates for small-scale biogas plants has not been conducted, despite these materials being the main input.

In this experimental study the heavy metal content of the biogas digestate from the anaerobic co-digestion of the mesophilic batch of poultry manure and sunflower hulls was investigated. The following conclusions can be drawn based on the experimental findings: there is an excess of Zn and Cu, which requires further adjustment of digestate doses and possible pretreatment to fix these metals in insoluble form to reduce their migratory activity and absorption by the root systems of plants.

In addition to their natural occurrence, heavy metals can enter the ecological system through anthropogenic activities such as sewage sludge disposal, application of pesticides and organic fertilizers, as well as by atmospheric deposition.^{95,96}

The relative solubility of heavy metals under physiological conditions determines the differences in biological significance and toxicity of heavy metals with respect to affinity to the environment and other interactions with macronutrients in the composition of living organisms. The low solubility of the trivalent or quadrivalent cations of Sn, Xie, Ga, Zr, and Th are not biologically significant. However, Fe, Mo and Mn are important trace elements with low toxicity; Zn, Ni, Si, V, Co, W and Cr are toxic elements with high to moderate importance as trace elements, and As, Ag, Sb, Cd, Hg, Pb and U have no useful function but are considered toxins to cells.

In addition to their important catalytic role, heavy metal cations, by forming complexes with biomolecules in complex biochemical reactions such as nitrogen fixation, water photolysis during photosynthesis, oxygen or nitrate respiration, one-electron catalysis, C—C-bond rearrangement,

urea hydrogen assimilation, and gene transcription in mRNA, can also form nonspecific complex compounds in cells in elevated concentrations, leading to toxic effects. Heavy metal cations such as Hg^{2+} , Cd^{2+} , and Ag^+ are very toxic and are often dangerous to any biological function.

Even the most important trace elements, such as Zn^{2+} or Ni^{2+} , and especially Si^{2+} , are toxic at elevated concentrations. All living forms have been forced to develop some kind of homeostasis system to ensure tight control of the concentration of heavy metal ions inside cells.

Figure 4 shows the potential forms of heavy metals in the anaerobic digestion substrate.

The presence of heavy metal ions (ferrous, chromium, copper, and zinc) in the environment, depending on their amount, has both a positive and negative effect on biogas production by anaerobic microbial association. The yield of biogas and methane depends on the composition of the compound in the form of which the metal ion is contained; complexing agents and other compounds present in the medium that change the bioavailability of ions; and the preliminary adaptation of the microbial association to the increased content of metal ions in the medium. Different anaerobic microorganisms and their associations tolerate specific concentrations of heavy metal ions without adversely affecting biogas production. Acceptable ranges include ferrous ions $4\text{--}150\text{ mg}\cdot\text{L}^{-1}$, oxides up to $10\text{ g}\cdot\text{L}^{-1}$, Cr^{3+} $28\text{--}50\text{ mg}\cdot\text{L}^{-1}$, Cu^{2+} $0.3\text{--}5\text{ mg}\cdot\text{L}^{-1}$, and Zn^{2+} $5\text{--}20\text{ mg}\cdot\text{L}^{-1}$.⁹⁷

Essential nutrients in digestate

The digestate also contains essential nutrients,⁹⁸ such as phosphorus (P), calcium (Ca), magnesium (Mg), potassium (K) and sodium (Na). We therefore analyzed the amount of those nutrients in the digestate to obtain precise results. Table 5 and Supporting Information, Appendix S1

(Supporting Information, Table S3) provide the results for K, P, Ca and Mg.

Digestate can be used as a fertilizer. However, it should contain minerals and organic ingredients in doses that are suitable for plants.

This study found that the digestate fractions analyzed were sourced from different biogas plants, which resulted in variations in the concentrations of specific biogenic elements, such as calcium, sodium, and phosphorus *etc.* The influence of the feedstock used and the technology of the biogas plant can also be important for digestate, as confirmed by Czekala⁹⁹.

The application of digestate as a fertilizer positively influences the content of macronutrients in the soil. The study by Koszel and Lorencowicz¹⁰⁰ observed a significant increase in potassium (+127.78%) and phosphorus (+18.79%) concentrations after digestate application. These elements are essential for maintaining water balance, photosynthesis, and plant growth. A slight increase in magnesium levels (+2.99%) and a stable pH also indicate improved conditions for nutrient absorption, making digestate an effective organic fertilizer for enhancing crop yields. Similar results were obtained by Mukhtar *et al.*,¹⁰¹ who found that the available phosphorus and potassium content increased from 4.28 to

Table 5. Essential nutrients detected in digestate (N = 30).

	K $\text{mg}\cdot\text{kg}^{-1}$	P $\text{mg}\cdot\text{kg}^{-1}$	Ca $\text{mg}\cdot\text{L}^{-1}$	Mg $\text{mg}\cdot\text{L}^{-1}$
Average	7981.4	63650.1	833.1	489.6
Min.	1690.0	8056.0	345.0	37.4
Max.	66475.0	520332.0	2154.0	1440.0
Frequent value range (Q1–Q3)	5258– 7365	27980– 76137	569–1014	125–702

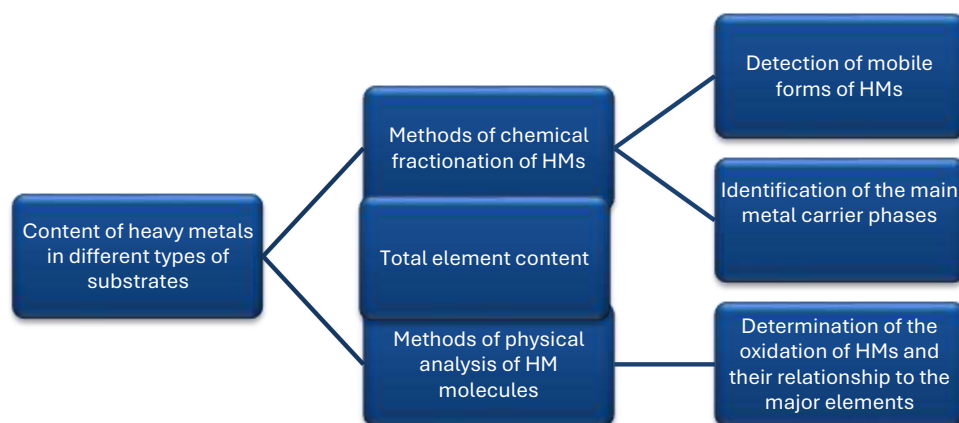


Figure 4. Forms of heavy metals in the substrate for anaerobic digestion and methods of detection.

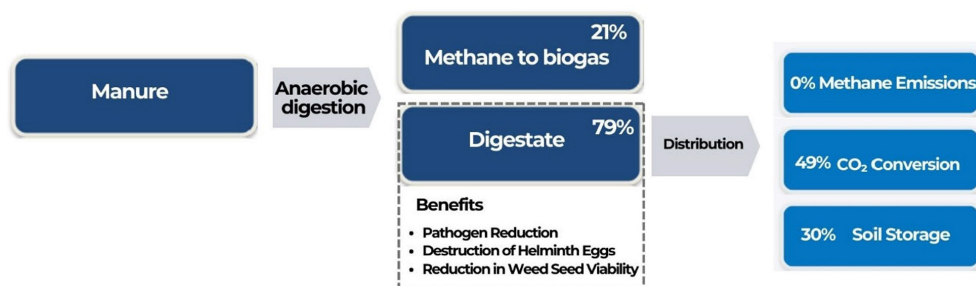


Figure 5. Summary of the directions of influence of digestate under the implementation of biogas technologies.

7.36 mg·kg⁻¹ and from 121.83 to 169.8 mg·kg⁻¹, respectively, after digestate application.

The effective use of digestate as an organic fertilizer can enhance crop yields and improve nutrient absorption conditions for plants. It supports the potential benefits of implementing biogas technologies, which include reducing environmental impacts associated with manure waste management and improving agricultural productivity. Figure 5 provides a complete summary of the environmentally friendly effects of the implementation of biogas technologies.

Conclusions

There is an urgent need for cost-effective methods to address environmental issues, especially pig and human manure, by reducing microbial and chemical contaminants. Experimental data related to the evaluation of digestate quality and several digestate properties were quantified and compared with aggregate data to investigate the suitability of different methods of increasing its value as a biofertilizer. Anaerobic digestion is an effective way to implement regenerative treatment and recovery of resources and the use of agricultural waste in Vietnam but gives rise to the important problem of the need for the complete disinfection of pathogenic bacteria. Pathogens have mechanisms to overcome adverse conditions (antimicrobial resistance, etc.) to survive to the greatest extent, providing spore-forming pathogens and VBNC pathogens as the main forms. This should be considered when applying the mesogenic digestion regimen.

It should be noted that long retention times and high ammonia concentrations have a positive effect on pathogen inactivation. These conditions can be achieved during anaerobic digestion and can be used for deep pathogen inactivation.

Using experimental data and analysis of previous studies, we developed a visualization model illustrating the decontamination of organic waste through anaerobic digestion under mesophilic conditions, considering

digestate quality indicators. The analysis of the composition of compensation tanks in small-scale biogas plants in Vietnam provided data indicating the potential use of digestate as a fertilizer due to its content of nitrogen compounds, phosphorus, potassium, magnesium, and other macronutrients. However, concerns must be addressed regarding elevated levels of certain heavy metals, particularly Cu, which exceeded acceptable limits in 87% of cases, and zinc in 54% of cases, potentially posing limitations for agricultural applications.

Further research is needed to assess feedstock quality, as it affects digestate composition directly. A deeper understanding of feedstock variability could provide valuable insights for optimizing digestate use in sustainable agriculture in Vietnam. This paper presented a comprehensive analysis of the environmental impact of biogas technology in a block diagram, integrating various points of view and approaches regarding digestate effects in biogas implementation.

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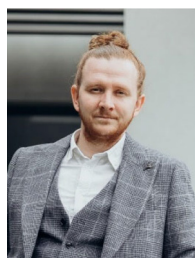
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Hynek Roubík

Associate Professor Dr Hynek Roubík is the director of the BioResources and Technology Division (BRT) and dean of the Faculty of Tropical AgriSciences (FTA) at the Czech University of Life Sciences Prague (CZU), Czech Republic. His

research focuses on biogas, waste management, and environmental engineering.



Viktoriia Chubur

Dr Viktoriia Chubur is a postdoctoral researcher in the BioResources and Technology Division (BRT) of the Czech University of Life Sciences Prague (CZU), Czech Republic. Her research focuses on anaerobic waste digestion for biogas production, ecology, environmental protection, and waste management.



Yelizaveta Chernysh

Dr Yelizaveta Chernysh is a senior researcher in the BioResources and Technology Division (BRT) of the Czech University of Life Sciences Prague (CZU), Czech Republic, and an associate professor at Sumy State University, Ukraine. Her research focuses on interdisciplinary research in waste recycling, environmental protection, bioenergy production, and sustainable feedstock.



Marek Jelínek

Marek Jelínek is a junior researcher and PhD candidate in the BioResources and Technology Division (BRT) of the Czech University of Life Sciences Prague (CZU), Czech Republic. His research focuses on sustainable rural development, waste management, and biogas technologies. He specializes in impact assessment and the valorization of organic waste for renewable energy production.



Le Dinh Phung

Dr Le Dinh Phung is a professor and Vice-Rector for Scientific Management and International Relations at Hue University of Agriculture and Forestry, Vietnam. His research specializes in livestock production and environmental management, with a focus on reducing ammonia, odor, and greenhouse gas emissions from livestock farming.

**Dinh Van Dung**

Associate Professor Dinh Van Dung is a researcher and lecturer in the Faculty of Animal Sciences and Veterinary Medicine, Hue University of Agriculture and Forestry, Vietnam. He has 18 years of research experience, his main areas of interest being animal nutrition and the quality of livestock products.

**Van Hau Duong**

Van Hau Duong is a junior researcher and PhD candidate in the BioResources and Technology Division (BRT) of the Czech University of Life Sciences Prague (CZU), Czech Republic. His research focuses on biogas technology, digestate processing, and the production of biochar for use as biofertilizers, soil amendments, and bioadsorbents for environmental remediation.

**Jan Banout**

Professor Jan Banout is a university professor and senior researcher in food security and sustainable agriculture practices in developing countries, with a particular interest in food and organic waste processing technology. He is also a senior researcher in the BioResources and Technology Division (BRT) of the Czech University of Life Sciences Prague (CZU), Czech Republic. His research interests include efficient agricultural practices, organic waste processing technologies, food security, and research for development (RforD).